Week 5, Lecture 2
Open quantum systems

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Walls & Milburn, Quantum Optics, 2nd Edition, Chapter 6

What you will learn today…

How coupling to an optical environment produces both damping and quantum noise.

Derive and interpret the quantum Langevin equation for a damped cavity mode.

Connect the Heisenberg-picture Langevin equation to the Lindblad master equation for the reduced system state.

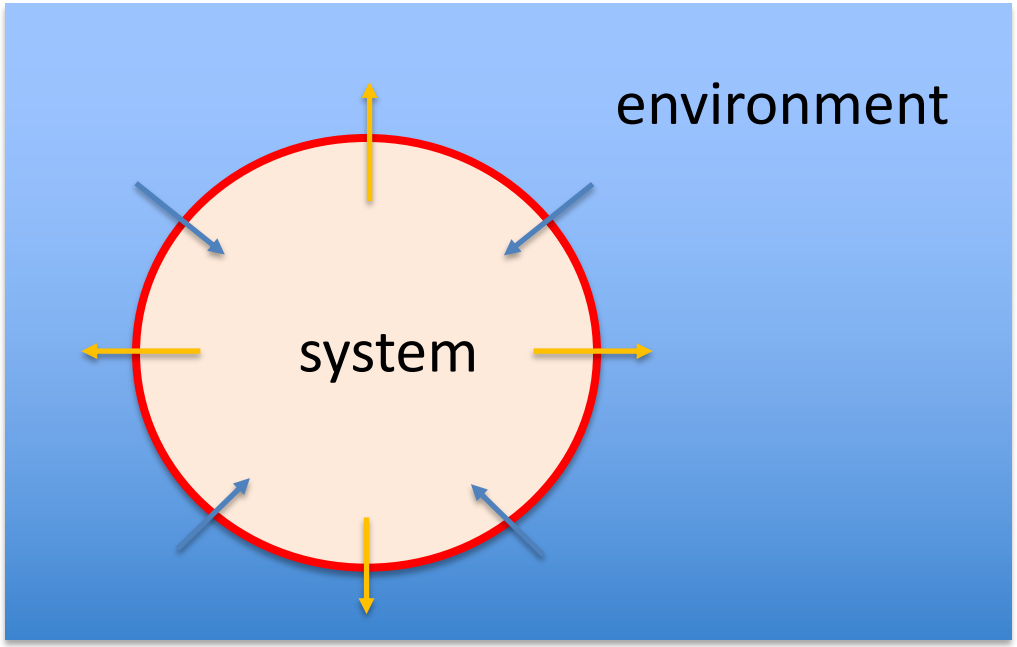
From Quantum Channels to Equations of Motion

Last lecture: What does the environment do?

A quantum channel describes the reduced input–output evolution of the system:

$$\hat{\rho}' = \mathrm{Tr}_E \left\{ \hat{U} \left(\hat{\rho} \otimes \hat{\rho}_E \right) \hat{U}^\dagger \right\}$$

- Loss channels exchange excitations with the environment.
- Phase damping suppresses coherence without exchanging energy.
- The interaction and the environment state determine the resulting channel.



This lecture: How does this evolution develop in time?

We model the environment as a continuum of bosonic modes and begin with

$$\hat{H} = \hat{H}_{sys} + \hat{H}_B + \hat{H}_i$$

Central goal: Derive the continuous-time dynamics of a quantum channel from a microscopic system–environment interaction.

Two descriptions of open-system dynamics

A localized quantum system interacts with a bath consisting of many simple harmonic oscillators, such as free optical modes.

- **System operator: $\hat{c}$**
- **Bath-mode operators :$\widehat{b_\omega}$**

Two equivalent descriptions

Picture	What evolves?	Equation of motion
Heisenberg picture	System operators, such as $\hat{c}(t)$	Quantum Langevin equation
Schrödinger picture	System density operator, $\hat{\rho}(t)$	Master equation

Both descriptions arise from the same system-bath Hamiltonian and give the same physical predictions.

Gardiner–Collett input–output theory

C. W. Gardiner and M. J. Collett
Input and output in damped quantum systems: Quantum stochastic differential equations and the master equation
Physical Review A 31, 3761–3774 (1985)

The derivation will:

- Model the environment as a continuum of bosonic field modes.
- Derive damping and environmental noise from the system–bath Hamiltonian.
- Identify the incoming field as the Langevin noise operator.
- Produce a quantum Langevin equation for system observables.
- Apply the result to a leaking optical cavity.



PHYSICAL REVIEW A VOLUME 31, NUMBER 6 JUNE 1985

Input and output in damped quantum systems: Quantum stochastic differential equations and the master equation

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(Received 29 October 1984)

We develop a formulation of quantum damping theory in which the explicit nature of inputs from a heat bath, and of outputs into it, is taken into account. Quantum Langevin equations are developed, in which the Langevin forces are the field operators corresponding to the input modes. Time-reversed equations exist in which the Langevin forces are the output modes, and the sign of damping is reversed. Causality and boundary conditions relating inputs to system variables are developed. The concept of "quantum white noise" is formulated, and the formal relationship between quantum Langevin equations and quantum stochastic differential equations (SDE's) is established. In analogy to the classical formulation, there are two kinds of SDE's: the Ito and the Stratonovich forms. Rules are developed for converting from one to the other. These rules depend on the nature of the quantum white noise, which may be squeezed. The SDE's developed are shown to be exactly equivalent to quantum master equations, and rules are developed for computing multitime-ordered correlation functions with use of the appropriate master equation. With use of the causality and boundary conditions, the relationship between correlation functions of the output and those of the system and the input is developed. It is possible to calculate what kind of output statistics result, provided that one knows the input statistics and provided that one can compute the system correlation functions.

I. INTRODUCTION

The treatment of inputs and outputs in quantum systems has a long history. In the context of quantum field theory, the most explicit formulation has been that of Lehman, Symanzik, and Zimmerman[1] (known as LSZ), in which operators for "in" and "out" quantum fields are formulated, and the S matrix derived as the unitary transformation relating the in and out operators. The matrix elements of this S matrix are then the scattering amplitudes, measurable by scattering experiments. The S-matrix elements are related by a "reduction formula" to the time-ordered products of vacuum expectation values of field operators. There is a range of approximate methods of tackling the computation of S-matrix elements, with differing degrees of simplicity and reliability.

In the theory of quantum amplification, we are most interested in how a well-defined subsystem, the amplifier, relates the output to the input. Caves[2] took the S-matrix point of view, and defined a theory of linear amplifiers, in which the S matrix related the output fields linearly to the input fields, and thus, without having to consider any of the details of the amplifying system, was able to derive fundamental limits on amplification for such linear amplifiers.

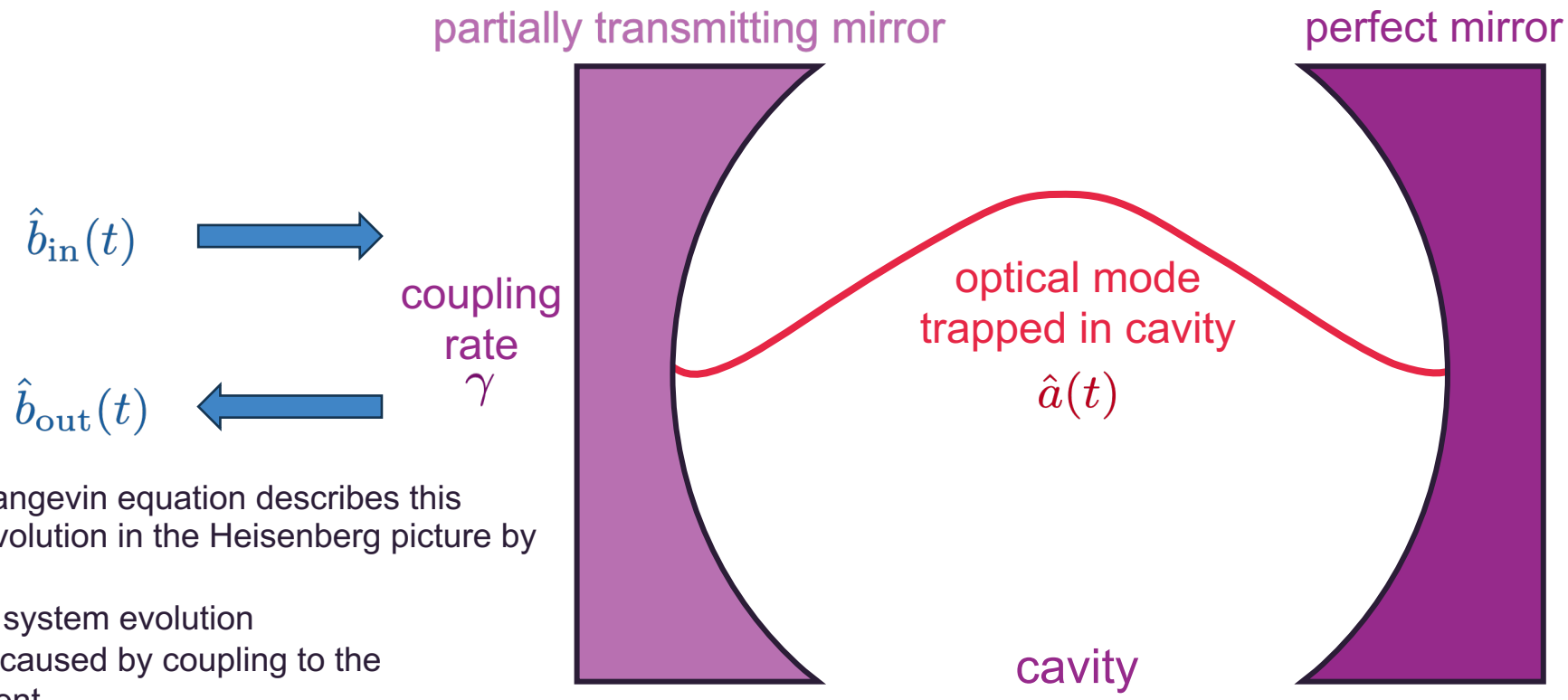
As far as quantum optics is concerned, these points of view are rather far from that normally used. The amplifying device itself can usually be described quite well, but the computation of inputs and outputs is rather poorly formulated. One normally thinks of the system as being in contact with a heat bath, which most usually consists of the vacuum modes of the electromagnetic field. By eliminating these modes, one obtains a damped equation of motion for the system, to which are added quantum noise terms. If required an additional classical deriving field is usually added separately, though it is clearly the result of one or more of the modes of the electromagnetic field being in a coherent state, rather than the vacuum. The formal equivalence of a bath mode being in a coherent state, and the addition of a classical driving field, is both rigorous and well known, and so in this case this procedure is merely *aesthetically* unsatisfactory in its separation of two different aspects of the electromagnetic field. However, no procedure has been developed so far which can cope adequately with the possibility of input fields which are neither coherent nor thermal, for example, squeezed fields, which are now of increasing interest.

The output modes are calculated usually by particular methods; e.g., in resonance fluorescence the radiated electromagnetic field is calculated from the solutions of Maxwell's equations,[3,4] while in the case of cavities linear propagation equations can be used to relate in operators to out operators.[5] Yurke and Denker[6] in their formulation of quantum circuit theory have demonstrated the intimate connection between the noise in conventional damped Heisenberg equations of motion, and the *input* operators for an infinite transmission line connected to a resonant circuit. As well as this, they showed that the boundary condition between the infinite transmission line and the finite resonant circuit gives rise to a time-reversal phenomenon, in which the damping changes sign, and the system is driven by the output operators.

This paper is devoted to the formulation of an idealized "quantum white-noise" method of looking at quantum systems. In Sec. II we formulate our system in terms of a somewhat idealized class of Hamiltonians, in which a fi-

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Single-sided cavity



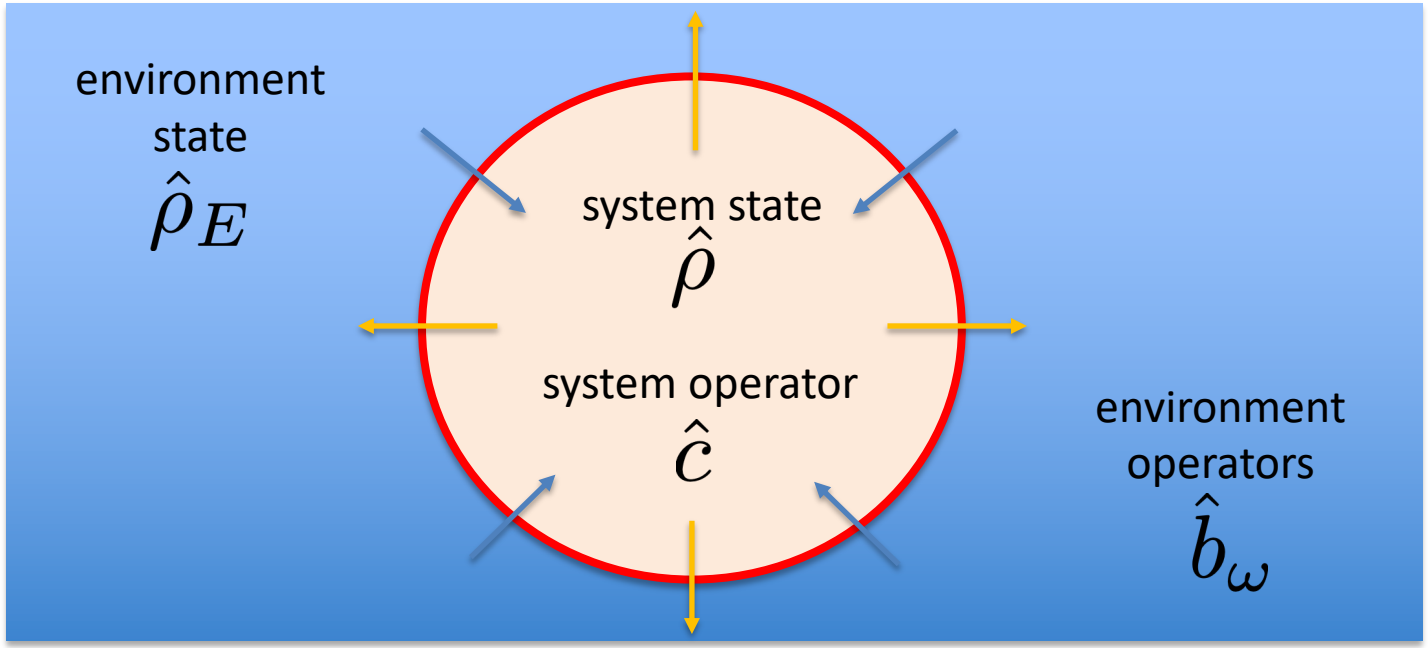
A quantum Langevin equation describes this continuous evolution in the Heisenberg picture by combining:

- Coherent system evolution
- Damping caused by coupling to the environment
- Quantum fluctuations entering from the environment

For the damped-cavity example, we will derive

$$\dot{\hat{a}} = -\frac{i}{\hbar}[\hat{a}, \hat{H}_{sys}] - \frac{\gamma}{2}\hat{a} + \sqrt{\gamma}\,\hat{b}_{\rm in}(t)$$

Model the System and Its Environment



The environment is modelled as a continuum of independent bosonic modes labelled by their angular frequency ω.

The total Hamiltonian is

$$\hat{H} = \hat{H}_{\text{sys}} + \hat{H}_B + \hat{H}_i$$

The free Hamiltonian of the bosonic bath is

$$\hat{H}_B = \hbar \int_0^\infty d\omega\, \omega\, \hat{b}_\omega^\dagger \hat{b}_\omega$$

The system-bath interaction is

$$\hat{H}_i = i\hbar \int_0^\infty d\omega\, \kappa(\omega) \left(\hat{b}_\omega^\dagger \hat{c} - \hat{c}^\dagger \hat{b}_\omega \right)$$

$\hat{c}$: system coupling operator
$\hat{b}_\omega$: annihilation operator for the bath mode at frequency ω
$\kappa(\omega)$: frequency-dependent coupling strength

$$\left[\hat{b}_\omega, \hat{b}_{\omega'}^\dagger \right] = \delta(\omega - \omega')$$

Approximations for an Optical Heat Bath

Only environmental modes close to the cavity resonance contribute significantly to real photon emission, absorption and cavity damping. $\omega \approx \omega_c$

1. Extend the frequency range

To simplify the derivation, formally extend the bath integral:

$$\int_0^\infty d\omega \longrightarrow \int_{-\infty}^\infty d\omega$$

The operators $\hat{b}_\omega$ for ω<0 represent fictitious modes introduced by this mathematical extension. They do not change the physical near-resonant dynamics.

2. Make the rotating-wave approximation

Expanding the full interaction gives energy-exchange and counter-rotating terms:

$$\left(\hat{b}_\omega^\dagger - \hat{b}_\omega\right)\left(\hat{c} + \hat{c}^\dagger\right) = \underbrace{\hat{b}_\omega^\dagger \hat{c} - \hat{c}^\dagger \hat{b}_\omega}_{\text{energy exchange}} + \underbrace{\hat{b}_\omega^\dagger \hat{c}^\dagger - \hat{b}_\omega \hat{c}}_{\text{counter-rotating}}$$

Only the slowly varying, excitation-exchange terms are retained.

$$\approx \hat{b}_\omega^\dagger \hat{c} - \hat{c}^\dagger \hat{b}_\omega$$

From the Hamiltonian to the Coupled Equations

Step 1: Start with the Heisenberg equation

For an operator with no explicit time dependence,

$$\dot{\hat{O}} = -\frac{i}{\hbar}[\hat{O}, \hat{H}]$$

Step 2: Apply it to one bath mode

Set $\hat{O} = \hat{b}_\omega$:

$$\dot{\hat{b}}_\omega = -\frac{i}{\hbar}[\hat{b}_\omega, \hat{H}_B] - \frac{i}{\hbar}[\hat{b}_\omega, \hat{H}_i] \qquad \left[\hat{b}_\omega, \hat{H}_{\text{sys}}\right] = 0$$

$$= \underbrace{-i\omega\hat{b}_\omega}_{\text{free bath evolution}} + \underbrace{\kappa(\omega)\hat{c}}_{\text{driven by the system}}$$

Step 3: Apply it to an arbitrary system operator

Because system operators commute with the bath Hamiltonian, $[\hat{a}, \hat{H}_B] = 0$

Therefore,

$$\dot{\hat{a}} = -\frac{i}{\hbar}[\hat{a}, \hat{H}_{sys}] + \int_{-\infty}^{\infty} d\omega\, \kappa(\omega) \left(\hat{b}_\omega^\dagger[\hat{a}, \hat{c}] - [\hat{a}, \hat{c}^\dagger]\hat{b}_\omega\right)$$

From the Hamiltonian to the Coupled Equations

Step 4: Assume a flat bath coupling

$$\kappa(\omega) \approx \sqrt{\frac{\gamma}{2\pi}}$$

Coupling constant is independent of frequency

Next step: Solve the bath equation and substitute it into the system equation.

$$\dot{\hat{b}}_\omega(t) + i\omega\hat{b}_\omega(t) = \kappa(\omega)\hat{c}(t)$$

Solving a Driven First-Order Equation

Consider the general equation

$$\dot{x}(t) = \lambda x(t) + f(t)$$

where $x(t)$ is the unknown quantity, λ is a constant, and $f(t)$ is a driving term.
Define a new quantity using an integrating factor:

$$z(t) = e^{-\lambda t} x(t)$$

Its derivative is

$$\dot{z}(t) = e^{-\lambda t} \left[\dot{x}(t) - \lambda x(t)\right] = e^{-\lambda t} f(t)$$

Integrating from t_0 to t gives

$$x(t) = \underbrace{e^{\lambda(t-t_0)} x(t_0)}_{\text{free evolution}} + \underbrace{\int_{t_0}^{t} dt'\, e^{\lambda(t-t')} f(t')}_{\text{accumulated driving}}$$

Apply the General Solution to the Bath Mode

Begin with the bath equation:

$$\dot{\hat{b}}_\omega(t) + i\omega\hat{b}_\omega(t) = \kappa(\omega)\hat{c}(t)$$

Applying the general solution gives

$$x(t) \to \hat{b}_\omega(t), \qquad \lambda \to -i\omega, \qquad f(t) \to \kappa(\omega)\hat{c}(t)$$

$$\hat{b}_\omega(t) = \underbrace{e^{-i\omega(t-t_0)}\hat{b}_\omega(t_0)}_{\text{free bath evolution}} + \underbrace{\kappa(\omega)\int_{t_0}^{t} dt'\, e^{-i\omega(t-t')}\hat{c}(t')}_{\text{bath driven by the system}}$$

Substituting the Bath Solution: Input and Damping

Start with the system equation and the bath solution:

$$\dot{\hat{a}}(t) = -\frac{i}{\hbar}[\hat{a}(t), \hat{H}_{\text{sys}}] + \int_{-\infty}^{\infty} d\omega\, \kappa(\omega) \left\{ \hat{b}_\omega^\dagger(t)[\hat{a}(t), \hat{c}(t)] - [\hat{a}(t), \hat{c}^\dagger(t)]\hat{b}_\omega(t) \right\}$$

$$\hat{b}_\omega(t) = e^{-i\omega(t-t_0)}\hat{b}_\omega(t_0) + \kappa(\omega) \int_{t_0}^{t} dt'\, e^{-i\omega(t-t')}\hat{c}(t'),$$

$$\hat{b}_\omega^\dagger(t) = e^{i\omega(t-t_0)}\hat{b}_\omega^\dagger(t_0) + \kappa(\omega) \int_{t_0}^{t} dt'\, e^{i\omega(t-t')}\hat{c}^\dagger(t')$$

Substitute both bath solutions into the system equation:

$$\begin{aligned}\dot{\hat{a}}(t) = & -\frac{i}{\hbar}[\hat{a}(t), \hat{H}_{\text{sys}}] \\ & + \underbrace{\int_{-\infty}^{\infty} d\omega\, \kappa(\omega) \left\{ e^{i\omega(t-t_0)}\hat{b}_\omega^\dagger(t_0)[\hat{a}(t), \hat{c}(t)] - [\hat{a}(t), \hat{c}^\dagger(t)]e^{-i\omega(t-t_0)}\hat{b}_\omega(t_0) \right\}}_{\text{initial bath field}} \\ & + \underbrace{\int_{-\infty}^{\infty} d\omega\, \kappa^2(\omega) \int_{t_0}^{t} dt' \left\{ e^{i\omega(t-t')}\hat{c}^\dagger(t')[\hat{a}(t), \hat{c}(t)] - [\hat{a}(t), \hat{c}^\dagger(t)]e^{-i\omega(t-t')}\hat{c}(t') \right\}}_{\text{system-driven bath response}}\end{aligned}$$

The Bath Initial Condition Becomes the Input Field

The first term in the solution for $\hat{b}_\omega(t)$ is

$$e^{-i\omega(t-t_0)}\hat{b}_\omega(t_0)$$

$$\hat{b}_\omega(t) = \underbrace{e^{-i\omega(t-t_0)}\hat{b}_\omega(t_0)}_{\text{free bath evolution}} + \underbrace{\kappa(\omega)\int_{t_0}^{t} dt'\, e^{-i\omega(t-t')}\hat{c}(t')}_{\text{bath driven by the system}}$$

Combining this term over all bath frequencies defines the input field:

$$\hat{b}_{\rm in}(t) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} d\omega\, e^{-i\omega(t-t_0)}\hat{b}_\omega(t_0)$$

$$\kappa(\omega) \approx \sqrt{\frac{\gamma}{2\pi}}$$

we obtain

$$\int_{-\infty}^{\infty} d\omega\, \kappa(\omega) e^{-i\omega(t-t_0)}\hat{b}_\omega(t_0) = \sqrt{\gamma}\,\hat{b}_{\rm in}(t)$$

The corresponding contribution to the system equation is

$$\sqrt{\gamma}\left(\hat{b}_{\rm in}^\dagger(t)[\hat{a},\hat{c}] - [\hat{a},\hat{c}^\dagger]\hat{b}_{\rm in}(t)\right)$$

Substituting the Bath Solution: Input and Damping

$$\dot{\hat{a}}(t) = -\frac{i}{\hbar}[\hat{a}(t), \hat{H}_{\rm sys}]$$
$$+ \underbrace{\int_{-\infty}^{\infty} d\omega\, \kappa(\omega) \left\{ e^{i\omega(t-t_0)} \hat{b}_\omega^\dagger(t_0)[\hat{a}(t), \hat{c}(t)] - [\hat{a}(t), \hat{c}^\dagger(t)] e^{-i\omega(t-t_0)} \hat{b}_\omega(t_0) \right\}}_{\text{initial bath field}}$$
$$+ \underbrace{\int_{-\infty}^{\infty} d\omega\, \kappa^2(\omega) \int_{t_0}^{t} dt' \left\{ e^{i\omega(t-t')} \hat{c}^\dagger(t')[\hat{a}(t), \hat{c}(t)] - [\hat{a}(t), \hat{c}^\dagger(t)] e^{-i\omega(t-t')} \hat{c}(t') \right\}}_{\text{system-driven bath response}}$$

For the system-driven term, use

$$\kappa^2(\omega) = \frac{\gamma}{2\pi}, \qquad \int_{-\infty}^{\infty} d\omega\, e^{\pm i\omega(t-t')} = 2\pi\delta(t-t'), \qquad \int_{t_0}^{t} dt'\, \delta(t-t') F(t') = \frac{1}{2} F(t)$$

Therefore, the system-driven bath response becomes

$$\boxed{\frac{\gamma}{2} \left\{ \hat{c}^\dagger(t)[\hat{a}(t), \hat{c}(t)] - [\hat{a}(t), \hat{c}^\dagger(t)] \hat{c}(t) \right\}}$$

Combining the Terms Gives the Quantum Langevin Equation

The input-field contribution is

$$\sqrt{\gamma}\left\{\hat{b}_{\rm in}^{\dagger}(t)[\hat{a}(t),\hat{c}(t)]-[\hat{a}(t),\hat{c}^{\dagger}(t)]\hat{b}_{\rm in}(t)\right\}$$

The damping contribution is

$$\frac{\gamma}{2}\left\{\hat{c}^{\dagger}(t)[\hat{a}(t),\hat{c}(t)]-[\hat{a}(t),\hat{c}^{\dagger}(t)]\hat{c}(t)\right\}$$

Combining these with the system Hamiltonian gives

$$\begin{aligned}\dot{\hat{a}}(t) = & -\frac{i}{\hbar}[\hat{a}(t),\hat{H}_{\rm sys}] \\ & +\left[\frac{\gamma}{2}\hat{c}^{\dagger}(t)+\sqrt{\gamma}\,\hat{b}_{\rm in}^{\dagger}(t)\right][\hat{a}(t),\hat{c}(t)] \\ & -[\hat{a}(t),\hat{c}^{\dagger}(t)]\left[\frac{\gamma}{2}\hat{c}(t)+\sqrt{\gamma}\,\hat{b}_{\rm in}(t)\right]\end{aligned}$$

The input field satisfies

$$\left[\hat{b}_{\rm in}(t),\hat{b}_{\rm in}^{\dagger}(t')\right]=\delta(t-t')$$

The Bath Appears as Damping and Input Noise

We solved the equations for the individual bath modes $\hat{b}_\omega(t)$ and substituted their solutions into the system equation.

The result is the quantum Langevin equation:

$$\dot{\hat{a}}(t) = \underbrace{-\frac{i}{\hbar}[\hat{a}(t), \hat{H}_{\text{sys}}]}_{\text{isolated system evolution}} + \underbrace{\frac{\gamma}{2}\left\{\hat{c}^\dagger(t)[\hat{a}(t), \hat{c}(t)] - [\hat{a}(t), \hat{c}^\dagger(t)]\hat{c}(t)\right\}}_{\text{damping from emission into the bath}} + \underbrace{\sqrt{\gamma}\left\{\hat{b}^\dagger_{\text{in}}(t)[\hat{a}(t), \hat{c}(t)] - [\hat{a}(t), \hat{c}^\dagger(t)]\hat{b}_{\text{in}}(t)\right\}}_{\text{incoming field and quantum noise}}$$

$\hat{a}$: the system operator whose evolution we want to calculate
$\hat{c}$: system coupling operator
$\hat{b}_{in}(t)$: the field arriving from the environment

Example — A Damped Cavity Mode

For a cavity mode coupled to the bath,

$$\hat{H}_{\rm sys} = \hbar\omega_c \hat{c}^\dagger \hat{c}$$

To calculate the evolution of the cavity amplitude, choose $\hat{a} = \hat{c}$

The required commutators are $[\hat{c}, \hat{c}] = 0, \qquad [\hat{c}, \hat{c}^\dagger] = 1, \qquad [\hat{c}, \hat{H}_{\rm sys}] = \hbar\omega_c \hat{c}$

Substituting these into the quantum Langevin equation gives

$$\begin{aligned}\dot{\hat{c}}(t) = & -\frac{i}{\hbar}[\hat{c}(t), \hat{H}_{\rm sys}] \\ & + \left[\frac{\gamma}{2}\hat{c}^\dagger(t) + \sqrt{\gamma}\hat{b}^\dagger_{\rm in}(t)\right] \underbrace{[\hat{c}(t), \hat{c}(t)]}_{0} \\ & - \underbrace{[\hat{c}(t), \hat{c}^\dagger(t)]}_{1} \left[\frac{\gamma}{2}\hat{c}(t) + \sqrt{\gamma}\hat{b}_{\rm in}(t)\right]\end{aligned}$$

Therefore,

$$\boxed{\dot{\hat{c}}(t) = -\left(i\omega_c + \frac{\gamma}{2}\right)\hat{c}(t) - \sqrt{\gamma}\,\hat{b}_{\rm in}(t)}$$

Remove the Fast Cavity Oscillation

The cavity Langevin equation is

$$\dot{\hat{c}}(t) = -\left(i\omega_c + \frac{\gamma}{2}\right)\hat{c}(t) - \sqrt{\gamma}\,\hat{b}_{\rm in}(t)$$

Define slowly varying operators in a frame rotating at ω_c:

$$\hat{c}(t) = \hat{c}'(t)e^{-i\omega_c t}, \qquad \hat{b}_{\rm in}(t) = \hat{b}'_{\rm in}(t)e^{-i\omega_c t}$$

Differentiating the cavity operator gives

$$\dot{\hat{c}}(t) = e^{-i\omega_c t}\left[\dot{\hat{c}}'(t) - i\omega_c\hat{c}'(t)\right]$$

Substitution cancels the free oscillation:

$$\boxed{\dot{\hat{c}}'(t) = -\frac{\gamma}{2}\hat{c}'(t) - \sqrt{\gamma}\,\hat{b}'_{\rm in}(t)}$$

$$[\hat{c}'(t), \hat{c}'^{\dagger}(t)] = 1, \qquad [\hat{b}'_{\rm in}(t), \hat{b}'^{\dagger}_{\rm in}(t')] = \delta(t-t')$$

If the environment modes are white noise

Return to the arbitrary system operator $\hat{a}$. Define the noise part of the short-time interaction:

$$d\hat{G}_{\text{noise}} = \sqrt{\gamma}\left(\hat{c}\,d\hat{B}^{\dagger} - \hat{c}^{\dagger}d\hat{B}\right)$$

See Gardiner and Collett for details

Expanding the Heisenberg evolution to second order gives

$$d\hat{a} = -\frac{i}{\hbar}[\hat{a}, \hat{H}_{\text{sys}}]\,dt + [\hat{a}, d\hat{G}_{\text{noise}}] + \frac{1}{2}\left[[\hat{a}, d\hat{G}_{\text{noise}}], d\hat{G}_{\text{noise}}\right].$$

For thermal white noise,

$$d\hat{B} \sim \sqrt{dt}, \qquad d\hat{B}^{\dagger}d\hat{B} = \bar{N}\,dt, \qquad d\hat{B}\,d\hat{B}^{\dagger} = (\bar{N}+1)dt$$

Therefore, the second-order term survives: $\frac{1}{2}\left[[\hat{a}, d\hat{G}_{\text{noise}}], d\hat{G}_{\text{noise}}\right] = \frac{\gamma}{2}(\bar{N}+1)\left(\hat{c}^{\dagger}[\hat{a},\hat{c}] - [\hat{a},\hat{c}^{\dagger}]\hat{c}\right)dt + \frac{\gamma}{2}\bar{N}\left(\hat{c}[\hat{a},\hat{c}^{\dagger}] - [\hat{a},\hat{c}]\hat{c}^{\dagger}\right)dt.$

The first-order term gives explicit input noise; the second-order noise products generate emission and absorption drift.

Quantum Langevin Equation for a White-Noise Bath

For a broadband thermal bath with no temporal memory, the Langevin equation can be written in differential form:

$$
\begin{aligned}
d\hat{a} = & -\frac{i}{\hbar}[\hat{a}, \hat{H}_{\mathrm{sys}}]\, dt \\
& + \frac{\gamma}{2}(\bar{N}+1)\left(\hat{c}^\dagger[\hat{a},\hat{c}] - [\hat{a},\hat{c}^\dagger]\hat{c}\right) dt \\
& + \frac{\gamma}{2}\bar{N}\left(\hat{c}[\hat{a},\hat{c}^\dagger] - [\hat{a},\hat{c}]\hat{c}^\dagger\right) dt \\
& + \sqrt{\gamma}\left([\hat{a},\hat{c}]\, d\hat{B}^\dagger(t) - [\hat{a},\hat{c}^\dagger]\, d\hat{B}(t)\right).
\end{aligned}
$$

The noise increment represents the input field accumulated over dt:

$$d\hat{B}(t) = \hat{b}_{\mathrm{in}}(t)\, dt$$

The thermal white-noise correlations are

$$
\begin{aligned}
\left\langle \hat{b}^\dagger_{\mathrm{in}}(t)\hat{b}_{\mathrm{in}}(t') \right\rangle &= \bar{N}\,\delta(t-t'), \\
\left\langle \hat{b}_{\mathrm{in}}(t)\hat{b}^\dagger_{\mathrm{in}}(t') \right\rangle &= (\bar{N}+1)\delta(t-t').
\end{aligned}
$$

White noise with spectral density $\bar{N}$

The Master Equation Tracks the Reduced System State

We have derived an equation for an arbitrary system operator:

$$\text{Heisenberg picture:} \qquad \frac{d\hat{a}(t)}{dt} = \text{system evolution} + \text{damping} + \text{input noise}$$

We now want an equation for the state of the system alone:

$$\hat{\rho}_S(t) = \mathrm{Tr}_B[\hat{\rho}_{SB}(t)]$$

$$\text{Schrödinger picture:} \qquad \frac{d\hat{\rho}_S(t)}{dt} = ?$$

The two descriptions must predict the same expectation values:

$$\langle \hat{a} \rangle(t) = \mathrm{Tr}_S[\hat{a}\,\hat{\rho}_S(t)]$$

Cyclicity of the Trace Transfers the Evolution to $\hat{\rho}$

For a thermal bath with no coherent amplitude,

$$\left\langle d\hat{B}(t)\right\rangle_B = \left\langle d\hat{B}^\dagger(t)\right\rangle_B = 0$$

After averaging over the bath,

$$\langle \hat{a}\rangle = \mathrm{Tr}(\hat{a}\hat{\rho}), \qquad \frac{d}{dt}\langle \hat{a}\rangle = \mathrm{Tr}(\hat{a}\dot{\hat{\rho}})$$

Using cyclicity of the trace,

$$\mathrm{Tr}(\hat{X}\hat{Y}\hat{Z}) = \mathrm{Tr}(\hat{Z}\hat{X}\hat{Y})$$

The Hamiltonian contribution becomes

$$-\frac{i}{\hbar}\,\mathrm{Tr}\left([\hat{a}, \hat{H}_{\mathrm{sys}}]\hat{\rho}\right) = \frac{i}{\hbar}\,\mathrm{Tr}\left(\hat{a}[\hat{\rho}, \hat{H}_{\mathrm{sys}}]\right)$$

The emission contribution becomes

$$\frac{1}{2}\,\mathrm{Tr}\left[\left(\hat{c}^\dagger[\hat{a},\hat{c}] - [\hat{a},\hat{c}^\dagger]\hat{c}\right)\hat{\rho}\right]$$

$$= \mathrm{Tr}\left[\hat{a}\left(\hat{c}\hat{\rho}\hat{c}^\dagger - \frac{1}{2}\hat{c}^\dagger\hat{c}\hat{\rho} - \frac{1}{2}\hat{\rho}\hat{c}^\dagger\hat{c}\right)\right].$$

The Density Operator Obeys a Lindblad Master Equation

The corresponding Schrödinger-picture equation is

$$\dot{\hat{\rho}} = \frac{i}{\hbar}[\hat{\rho}, \hat{H}_{\rm sys}] + \frac{\gamma}{2}(\bar{N}+1)\left(2\hat{c}\hat{\rho}\hat{c}^\dagger - \hat{c}^\dagger\hat{c}\hat{\rho} - \hat{\rho}\hat{c}^\dagger\hat{c}\right) + \frac{\gamma}{2}\bar{N}\left(2\hat{c}^\dagger\hat{\rho}\hat{c} - \hat{c}\hat{c}^\dagger\hat{\rho} - \hat{\rho}\hat{c}\hat{c}^\dagger\right).$$

$\hat{c}$, with rate $\gamma(\bar{N}+1)$: emission into the bath
$\hat{c}^\dagger$, with rate $\gamma\bar{N}$: absorption from the bath

This is the Lindblad form for a system exchanging excitations with a thermal bath.

Summary

- The quantum Langevin equation is used to calculate how system operators evolve when both damping and environmental fluctuations matter—for example, the field inside a lossy optical cavity.
- For a thermal white-noise bath, second-order noise products generate the emission and absorption terms in the system dynamics.
- The quantum Langevin and Lindblad master equations describe the same open-system physics from the Heisenberg and Schrödinger pictures, respectively.